4 SMS Data Coding Scheme

((This subsection should really be moved to the SMS standard, 3GPP TS 23.040, Technical realization of the Short Message Service (SMS)))

Bits 7..4	***Bits 3..0 (sometimes 5..0)***		
00xx (0x00..0x3F)	**General Data Coding indication**		
	Bit 5, if set to 0, indicates the text is uncompressed		
	*Bit 5, if set to 1, indicates the text is compressed using the compression algorithm defined in 3GPP TS 23.042 [13] (**strongly deprecated;** 3GPP TS 23.042: **to withdraw**)*		
	Bit 4, if set to 0, indicates that bits 1 to 0 are reserved and have no message class meaning; interpret as Class 1.		
	Bit 4, if set to 1, indicates that bits 1 to 0 have a message class meaning.		
	Bit 3	Bit 2	Alphabet:
	0	0	**The 7-bit alphabet as set in the TP User Data Header (default alphabet ID is 0x00)**
	0	1	8 bit data
	1	0	UCS2/**UTF-16BE** (16 bit) [10]
	1	1	Reserved
	Bit 1	Bit 0	Message Class (if bit 4 is set to 1):
	0	0	Class 0: Immediate.
	0	1	Class 1: Default meaning: ME-specific.
	1	0	Class 2: (U)SIM specific message.
	1	1	Class 3: Default meaning: TE-specific (see 3GPP TS 27.005 [8])
01xx (0x40..0x7F)	**General Data Coding indication (as Group 00xx), but with Message Marked for Automatic Deletion**		
	Bits 5 to 0 are interpreted as for Group 00xx.		

	This group can be used by the SM originator to mark the message (stored in the ME or (U)SIM) for deletion after reading irrespective of the message class. The way the ME will process this deletion should be manufacturer specific but shall be done without the intervention of the End User or the targeted application. The mobile manufacturer may optionally provide a means for the user to prevent this automatic deletion.
1000..1011 (0x80..0xBF)	Reserved coding groups
1100 (0xC0..0xCF)	**Message Waiting Indication Group (7-bit alphabet as set in TP User Data Header): Class 0, Discard Message After Reading**
	The specification for this group is exactly the same as for Group 1101, except that: after presenting an indication and storing the status, the ME may discard the contents of the message.
	The ME shall be able to receive, process and acknowledge messages in this group, irrespective of memory availability for other types of short message.

1101 (0xD0..0xDF)	**Message Waiting Indication Group (7-bit alphabet as set in TP User Data header): Class 0 and 1, Store Message** This Group defines an indication to be provided to the user about the status of types of message waiting on systems connected to the GSM/UMTS PLMN. The ME should present this indication as an icon on the screen, or other MMI indication. The ME shall update the contents of the Message Waiting Indication Status on the SIM (see 3GPP TS 51.011 [18]) or USIM (see 3GPP TS 31.102 [17]) when present or otherwise should store the status in the ME. In case there are multiple records of EF_{MWIS} this information shall be stored within the first record. The contents of the Message Waiting Indication Status should control the ME indicator. For each indication supported, the mobile may provide storage for the Origination Address. The ME may take note of the Origination Address for messages in this group and group 1100. Text included in the user data is coded in the 7-bit **alphabet as set in the TP User Data Header**. Where a message is received with bits 7..4 set to 1101, the mobile shall store the text of the SMS message in addition to setting the indication. The indication setting should take place irrespective of memory availability to store the short message. Bits 3 indicates Indication Sense: Bit 3 0 Set Indication Inactive 1 Set Indication Active Bit 2 is reserved, and set to 0 Bit 1 Bit 0 Indication Type: 0 0 Voicemail Message Waiting 0 1 Fax Message Waiting 1 0 Electronic Mail Message Waiting 1 1 Other Message Waiting: Mobile manufacturers may implement this indication as an additional indication without specifying the meaning.

1110 (0xE0..0xEF)	**Message Waiting Indication Group (UCS2/UTF-16BE): Class 0 and 1, Store Message** The coding of bits 3..0 and functionality of this feature are the same as for the Message Waiting Indication Group above, (bits 7..4 set to 1101) with the exception that the text included in the user data is coded in the uncompressed UCS2/UTF-16BE character set.
1111 (0xF0..0xFF)	**????????? the 1111 group seems to be an editorial mistake** reserved
	Bit 3 Bit 2 Alphabet:
	0 0 **The 7-bit alphabet as set in the TP User Data Header (default alphabet ID is 0x00)**
	0 1 8 bit data
	1 0 Reserved (*UCS2/**UTF-16BE** (16 bit) [10]…)*
	1 1 Reserved
	Bit 1 Bit 0 Message Class:
	0 0 Class 0: Immediate.
	0 1 Class 1: Default meaning: ME-specific.
	1 0 Class 2: (U)SIM specific message.
	1 1 Class 3: Default meaning: TE-specific (see 3GPP TS 27.005 [8])